Ahsanul Hoque

Founder and CEO, Livora | Software Engineer (Full-Stack, AI and Embedded Systems)

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SUMMARY

Computer Science undergraduate (University of Chittagong, expected 2026) and Founder/CEO of Livora, a full-stack AI-powered healthcare platform that won Champion at the JnU AI and IT Fest Brainchild 2.0 and 2nd place with an incubation package at the ITBI Student Startup Pitch Fest 2026 (CUET). Ships production-grade full-stack and applied-AI systems across healthcare, e-commerce, embedded/IoT, and data engineering. Comfortable across the stack: React/TypeScript frontends, Node.js/Express APIs, SQL data modeling, LLM/RAG integration, and embedded systems (ESP32, LoRa).

EXPERIENCE

Founder and CEO - Livora (Self-employed) | May 2026 - Present
Chattogram, Bangladesh (Hybrid)

- Lead product, engineering, and business strategy for Livora, an AI-powered healthcare platform serving patients, doctors, lab technicians, and hospital administrators
- Won Champion at the JnU AI and IT Fest Brainchild 2.0 and 2nd place at the ITBI Student Startup Pitch Fest 2026 (CUET), the latter securing 6 months of free incubation space, mentoring, and startup-ecosystem support worth BDT 100,000
- Conducted an AI training workshop on the platform for practicing surgeons in Chittagong

SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL

Web and Backend: React, Node.js, Express, Prisma, Sequelize, REST APIs, Socket.IO

AI / LLM: LLM Integration (Google Gemini), RAG (Retrieval-Augmented Generation), Prompt Injection Defense, Langfuse (LLM Observability), OCR (Tesseract), Speech-to-Text (Deepgram)

Data and Infrastructure: PostgreSQL, MySQL, Redis, Docker, HL7 / FHIR / DICOM (healthcare interoperability), Cloudinary, Pino, Sentry

Embedded and IoT: ESP32, STM32, Arduino, FreeRTOS, LoRaWAN, Raspberry Pi, ThingSpeak

Other: Data Structures and Algorithms, Competitive Programming, Blockchain Fundamentals, Project Management

PROJECTS

Livora - AI-Powered Healthcare Platform | *Oct 2025 - Present*
React, TypeScript, Node.js, PostgreSQL, LLM/RAG, Socket.IO

- Building a full-stack clinical platform covering appointment booking, voice-driven prescriptions, lab order/result management with payments, DICOM/FHIR and HL7 ingestion, and multi-disciplinary team (MDT) video consultations
- Implemented a persistent AI memory system and a RAG-based health assistant with medicine safety checks, risk stratification, and prompt-injection defenses; instrumented with Langfuse and Sentry

Vision-Based Autonomous Navigation for a Quadrotor in GPS-Denied Environments | *Aug 2026 - Present*
Python, PyBullet, NumPy, A/RRT*, PID Control, IEEE LaTeX*

- Designed a full perception-planning-control pipeline for GPS-denied quadrotor navigation using only onboard depth-camera vision, simulated in PyBullet
- Benchmarked A* against RRT* across 60 trials (3 obstacle densities x 2 planners x 10 trials); A* won on success rate and path efficiency (1.03x optimal vs 1.06-1.11x) with about 30x lower planning latency (36ms vs approx. 1.1-1.2s)
- Wrote up the full study as a 5-page IEEE conference paper with experimental results and figures generated from the sweep data

Bengali Medical Dialogue Generation - Nascenia AI Hackathon | *Aug 2026*
Python, PyTorch, LoRA, NLP

- Fine-tuned an LLM under 3B parameters (LoRA on Qwen3-1.7B) to generate doctor-style responses in Bengali to patient questions for a Kaggle competition hosted by Nascenia
- Evaluated TF-IDF retrieval, retrieval+generation hybrids, semantic retrieval, and model-soup ensembling; documented what worked and why

Scraper Pipeline - AI-Powered Web Scraping and Extraction System | *Jul 2026 - Present*
Python, Playwright, Anthropic/OpenAI/Gemini/Groq/DeepSeek, pytest

- Built a 4-stage pipeline (discover, collect, extract, clean) with a cost-aware multi-tier LLM cascade and cross-provider fallback chain that only escalates to a costlier model when deterministic validation of the cheaper tier's output fails

- Implemented Cloudflare/WAF bypass via headless-browser fallback, crash-safe atomic writes, resume-by-default batch processing, and a 92-test no-network suite covering every pure-logic component

Second Brain - Autonomous AI Agent for Productivity and Automation | *May 2026 - Present*
Claude Code, Python, Telegram Bot API, Git, Cron
- Built an autonomous AI agent that runs study, startup, and daily-task workflows from a git-versioned Obsidian vault, with a Telegram bot as a mobile interface for logging progress, marking tasks done, and queuing background jobs
- Orchestrated 4 scheduled cron jobs (morning briefing, logging nudge, nightly re-planning, overnight task runner) atop a compiled knowledge-caching layer that processes source material once and serves every later query from cache

ResCris - Long-Range Emergency and Crisis Management Network | *2025*
ESP32, LoRa (SX1278), GPS, Embedded Systems
- Built an internet-independent emergency communication system using LoRaWAN for disaster response, with real-time SOS reporting, GPS-based tracking, and a multi-gateway architecture
- Presented at Chittagong Science Carnival 4.0 with a teammate

Aurora Gadgets - Full-Stack E-Commerce Platform | *Apr 2026 - May 2026*
React, TypeScript, Express, Prisma, MySQL, Docker
- Built a complete storefront and admin system covering products, categories, brands, cart, and checkout with category-based product attributes
- Containerized the application with Docker for reproducible deployment

Agri Supply Chain Transparency Platform | *2025*
Blockchain
- Designed a blockchain-based platform to make farm-to-consumer pricing transparent and verifiable, aimed at closing the margin gap between farmer income and consumer prices
- Runner-up, Hackathon Segment, Chittagong Science Carnival (hosted by CUSS)

Warehouse Management and Billing System | *Freelance*
Node.js, MySQL
- Built an inventory, billing, and sales-analytics system for a retail client

Sales Data Warehouse and BI Reporting | *Aug 2025 - Sep 2025*
SQL, Python, Jupyter Notebook
- Designed and built a modern SQL data warehouse from scratch, consolidating sales data via ETL and dimensional modeling
- Built interactive BI dashboards with Python (Pandas, Matplotlib, Seaborn) for trend and segmentation analysis

AgroEdge AI - Precision Agriculture Platform | *Jan 2026 - Apr 2026*
Python, Raspberry Pi, ThingSpeak
- Built an AI-first precision-agriculture stack: data pipeline, irrigation model training, edge inference, and a Raspberry Pi dashboard integrated with ThingSpeak

WaifOS - Multitasking OS for ESP32 | *Mar 2025 - May 2025*
C++, Arduino, FreeRTOS
- Built a lightweight multitasking operating system for ESP32 featuring an app menu, display rendering, and input handling
- Implemented WiFi connectivity and concurrent task scheduling on constrained embedded hardware

EDUCATION

University of Chittagong - B.Sc. in Computer Science and Engineering | 2022 - 2026 (Expected)
CGPA: 3.50/4.00 (through 6th semester)

Govt. City College, Chattogram - Higher Secondary Certificate, Science (GPA 5.00) | 2019 - 2022

Chittagong Collegiate School - Secondary School Certificate, Science (GPA 5.00) | 2013 - 2019

AWARDS AND ACHIEVEMENTS

- Champion, JnU AI and IT Fest Brainchild 2.0, 2026 - for Livora
- 2nd Place, ITBI Student Startup Pitch Fest 2026, CUET - for Livora (incl. incubation support)
- 1st Place, Robo Soccer, EEE Fest 2025, University of Chittagong
- Runner-up, Hackathon Segment, Chittagong Science Carnival (hosted by CUSS)
- 1st Runner-up, Robo Soccer, Bangabandhu Innovation Fair 2024, Premier University
- Finalist, Bangabandhu Innovation Grant (BIG), 2023
- Champion, ETE DecaHz 2022 - SDG-focused idea competition (algae-based circular economy model)
- 2nd Runner-up, Intra-Department Programming Contest, 22nd Batch

LEADERSHIP AND ACTIVITIES

- President, N-Cube (Science Club), Chittagong Collegiate School
- Joint Secretary, Debating Society, Chittagong Collegiate School